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Quantifying Skin Friction Drag on Tyres: Wind Tunnel Analysis

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BSc Advanced Technology

By

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Abstract

Estimating skin friction drag through indirect methods is typically used to characterize wall-bounded flow over rough surfaces. However, the difficulty of implementing such methods on complex geometries and irregular surfaces where scaling and similarity laws may not hold introduces a number of complications. The floating element sensor is a method that benefits from directly measuring and is not dependent on the veracity of correlating functions. This paper presents and tests a design that implements the floating element concept to investigate whether different surface roughnesses of a tyre's side panel influence skin friction drag. A numerical simulation using Reynolds-averaged Navier Stokes is also delivered in this paper to validate the experimental method through a comparative analysis. The turbulent flow over a smooth-wall flat plate at zero-incident is used as a benchmark to validate the two methods before studying the flow over the tyre's side panel. The results of this study indicate a further need for research into the uncertainties and sources of error surrounding this method, as a definitive conclusion could not be drawn.

Keywords: Skin friction drag, Wall-bounded turbulent boundary layer, floating element, Surface roughness, Wind tunnel analysis

Contents

Acronyms

APG adverse pressure gradient. 12, 16, 23, 27

CFD computational fluid dynamics. 6–8, 17, 18, 22, 34–37

CFL Courant-Friedrichs-Lewy. 18

CV control volume. 18, 19, 23

DNS direct numerical simulations. 13, 17, 21, 22

DOF degree of freedom. 27, 28, 36

FE floating element. 1, 26–30, 32, 36, 37

HWA hot-wire anemometry. 26, 36

LDV laser doppler velocimetry. 26

LES large eddy simulations. 17

RANS Reynolds-averaged Navier Stokes. 1, 8, 17, 18, 21, 35, 36

SIMPLE Semi-Implicit Method for Pressure-Linked Equations. 18

SNR signal to noise ratio. 30, 36

SST shear stress transport. 17

TBL turbulent boundary layer. 6, 9, 11–13, 19, 20, 22, 23

ZPG zero pressure gradient. 11–13, 16

Symbols

ΔS	First layer height [m]
ΔU^+	Roughness function [-]
δ^*	Displacement thickness [m]
δ_{99}	Boundary layer thickness [m]
κ	*von Karman* constant [-]
μ	Dynamic viscosity [Pa·s]
ν	Kinematic viscosity [m^2/s]
ρ	Density of air [kg/m^3]
τ	Shear stress [Pa]
τ_w	Wall shear stress [Pa]
$\mathbf{V}$	Velocity field [m/s]
θ	Momentum thickness [m]
A, B, C	Constants [-]
C_f	Coefficient of friction [-]
C_P	Coefficient of pressure [-]
D	Tyre diameter [m]
f_n	Natural frequency [Hz]
F_w	Measured force of wall [N]
k	Roughness height [m]
k	Stiffness [N/m]
k_f	Stiffness of flexure [N/m]
k_s	Equivalent sand-grain roughness [m]
k_s^+	Roughness Reynolds number [-]
$k_{\mathbf{rough}}$	Upper band of transitionally rough regime [-]
$k_{\mathbf{smooth}}$	Lower band of transitionally rough regime [-]
L	Characteristic length [m]
m	Mass of tyre [kg]
M_∞	Mach number [-]
P	Pressure [Pa]
P_∞	Freestream pressure [Pa]
P_w	Pressure at wall [Pa]
R	Reaction force [N]
Re	Reynolds number [-]
S	Surface area [m^2]
u'	Streamwise component of the fluctuating velocity [m/s]
U	Mean velocities [m/s]
u	Streamwise velocity component [m/s]
U^+	Dimensionless mean velocity [-]
U_∞	Freestream velocity [m/s]
U_τ	Friction velocity [m/s]
U_e	Velocity at boundary [m/s]
v'	Normal-wall component of the fluctuating velocity [m/s]
v	Normal-wall velocity component [m/s]
w'	Spanwise component of the fluctuating velocity [m/s]
w	Spanwise velocity component [m/s]
x	Streamwise direction [m]
y	Normal-wall direction [m]
y^+	Dimensionless wall normal-wall direction [-]
z	Spanwise direction [m]

1 Introduction

In automotive engineering, understanding the interaction of a vehicle's dynamics with the surrounding flow is paramount for enhancing performance. As manufacturers pursue innovation, they must ensure that new parts exposed to the flow improve or, at the very least, do not worsen the vehicle's performance. This paper investigates how adjustments in surface roughness may influence the amount of skin friction drag exerted on a body by looking at varying roughnesses in a car tyre's side panel.

1.1 Research framework

No matter the complex shape of a moving body, the aerodynamic resistance it experiences can be decomposed into two sources: pressure and skin friction drag [1, p. 19]. While George Gabriel Stokes laid the foundation for understanding viscous friction effects in fluid flow, it was the work of *Hagen* and *Darcy* that explored the impact of roughness on friction drag [2, p. 1]. Nowadays, Prandtl's boundary layer theory plays a crucial role in analysing the flow behaviour and is used in this study to examine causes of skin friction drag [3, p. 301].

Since then, intentionally manufactured roughness features have been employed in various aerodynamic applications to improve performance, reduce drag and enhance heat transfer [2, p. 2]. Sharkskin denticles on aircraft surfaces [4], [5] and dimples on golf balls [6], [7] are a few examples of drag reduction. Unintentional surface roughness from byproducts of manufacturing processes or surface degradation mechanisms, such as corrosion, erosion, and deposition processes, yields a similar result [8]. Experimental methods such as the work of Nikuradse, who developed the empirical expression for surface roughness employed in this study [9], are often implemented to observe the complicated interaction between the flow and rough surface.

1.2 Problem formulation

This research investigates the effects of various surface roughnesses of a tyre's side panel on skin friction drag. Therefore, the main research question is stated:

How can the skin friction drag exerted on a tyre's side panel in turbulent flow conditions be quantified for a variety of surface roughnesses?

Literature study and theoretical analysis of the flow case assist in exploring and addressing the following topics related to the research. Firstly, turbulence modelling of wall-bounded external flows is explored in depth to investigate how surface roughness impacts different fluid flow regions and alters flow characteristics such as velocity distribution. Special attention is placed on how the effect of surface roughness are modelled and what relation is expected as roughness is increased. Secondly, the tyre's side panel is curved and irregular, which poses how the drag components are expected to change with the roughness of such a geometry.

Once the theory behind the flow case is presented, the accuracy of a computational fluid dynamics (CFD) simulation made to represent the flow situation is questioned. Comparing it with the wind tunnel experiment designed to measure the skin friction drag is then used to conclude the study.

1.3 Premise of report

This paper examines the turbulent boundary layer (TBL) over the exposed side panel when the tyre is laid horizontally (Figure 1). The tyre is considered rotation-free, with a cover placed over its opening so that the side panel is the central area of interest. The airflow is in contact with the surface from the leading edge to the trailing edge across the tyre's entire surface area.

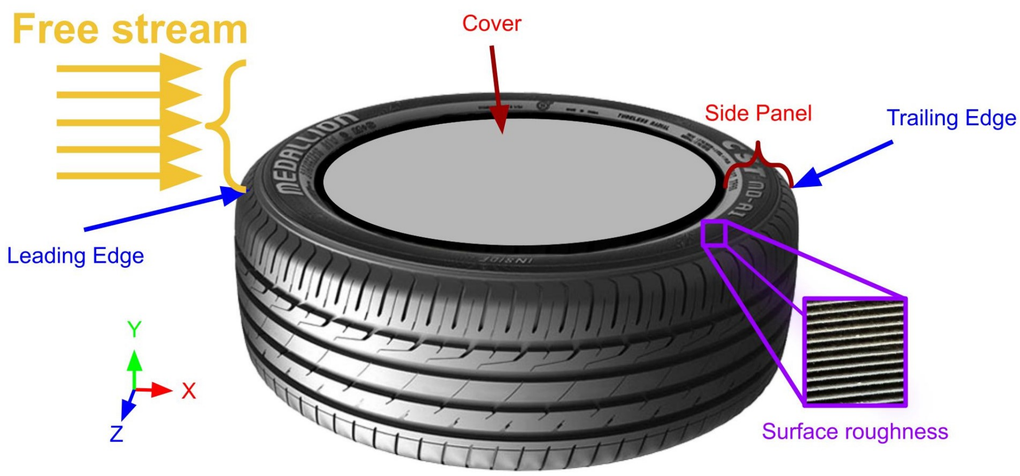


FIGURE 1: Flow case considered. Tyre image from [10]

Restricting the studied flow this way does not replicate the real-world example of a car driving down a road; that is not the purpose of this study. Instead, the flow case is limited to exploring whether different tyre materials, which consist of different roughnesses, significantly impact the exerted friction drag.

Four standard 24-inch (≈ 63 [cm]) car tyres with identical side panel profiles and different surface roughnesses are tested. Windspeeds ranging from 0 to 130 [km/h] chosen to represent standard driving speeds [11] are evaluated. For these speeds, the mach number M_∞ is lower than 0.3; therefore, the flow is assumed to be incompressible. Furthermore, the Reynolds number Re (equation 1) depicts the type of airflow, where the characteristic length L is the tyre's diameter. Reynolds number is the ratio of inertial to viscous forces within a fluid flow and depicts whether the flow is laminar or turbulent.

$$Re = \frac{\rho U L}{\mu} = \frac{\text{Inertial forces}}{\text{Viscous Forces}} \tag{1}$$

The remainder of this report presents the design of a wind tunnel measurement set-up aimed at quantifying skin friction drag, along with a CFD simulation to compare and validate the experiment. Firstly, the preliminary theoretical analysis of turbulence modelling and the consequences of surface roughness on flow behaviour is discussed in section 2. After that, the set-up and results of a CFD are shown in section 3 followed by the design process and measurement results of a wind tunnel experiment in section 4. The study concludes by comparing the implications of the two approaches and the study's limitations.

2 Theoretical Analysis

The velocity field $\mathbf{V} = (u, v, w)^T$ in this study has the components of u, v and w for the streamwise (x), wall-normal (y), and spanwise (z) directions (Figure 1), regarding the mean velocity magnitude being represented by U. The superscript "+" denotes dimensionless wall quantities; specificaly the velocity $U^+ = U/U_\tau$ and normal distance $y^+ = yU_\tau/\nu$. The term U_τ is the friction velocity given by the formula $U_\tau = \sqrt{\tau_w/\rho}$, and the flow case considered is an external incompressible flow in turbulent conditions governed by the Navier-Stokes continuity and momentum balances.

2.1 Turbulent boundary layer

Turbulence is characterised by a time-dependent three-dimensional situation in which fluid properties mix vigorously due to irregular fluctuations and swirling motions in velocity and pressure [12]. Critical features of turbulent flow include vorticity and eddying motion, which lead turbulent flow to be defined by a high Reynolds number. Hence implying that viscous forces are small compared to the inertial forces [13, p. 187].

The steady Reynolds-averaged Navier Stokes (RANS) equations for incompressible flow are used to explain the turbulent flow behaviour and the CFD method used in the coming section to study it. These RANS equations consist of the continuity and momentum (equations 2, 3); where i, j = 1, 2, and 3 represent the directions x, y and z, respectively.

$$\frac{\partial U_i}{\partial x_i} = 0 \tag{2}$$

$$\frac{\partial (U_i U_j)}{\partial x_j} = -\frac{1}{\rho}\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\nu\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) - \overline{u_i' u_j'}\right] \tag{3}$$

The Reynolds-averaging decomposes the chaotic flow into mean component U and fluctuating component u' such that $u_i = U + u_i'$. This process results in additional stress term $-\rho\overline{u_i' u_j'}$, from the nonlinear nature of the convective terms [3, p. 237-239]. These fluctuating components cannot be solved directly and must be expressed in terms of mean flow quantities. This is known as the turbulence closure problem.

Prandtl's boundary layer concept is used to analyse the behaviour of the flow so that the shear stress distributions for viscous flows can later be quantified [1, p. 997-1003]. Flow over a smooth flat plate (Figure 2(a)), illustrets the inner and outer velocity boundary layers and the velocity profile $U(y)$.

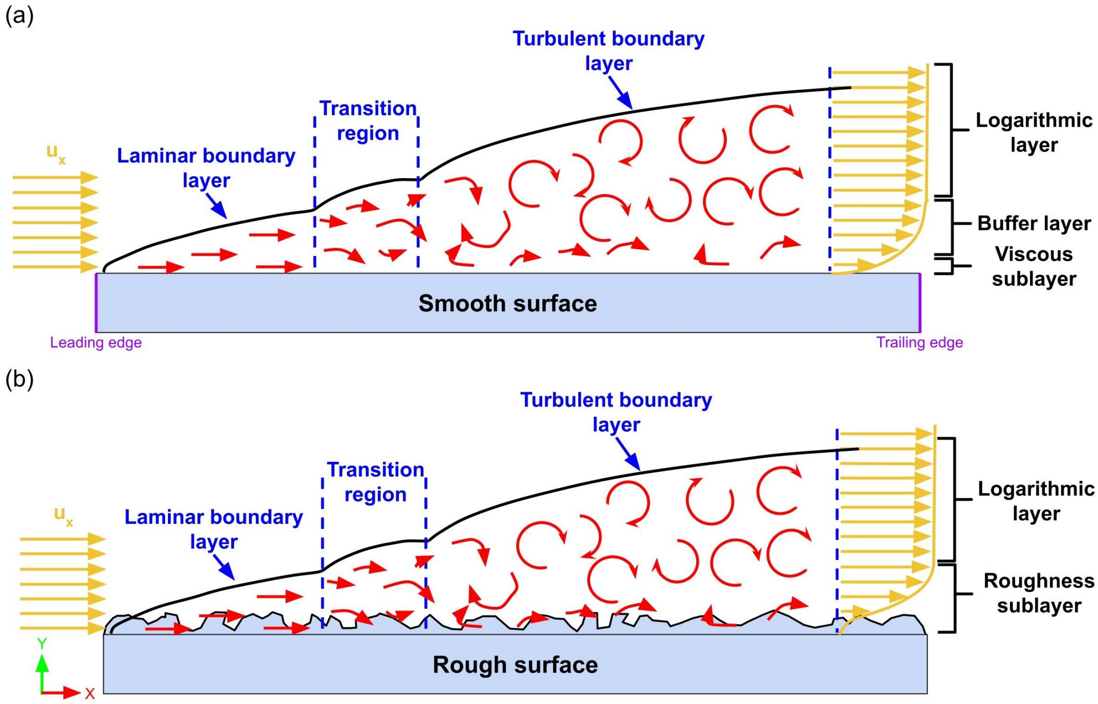


FIGURE 2: Flow boundary layer for (a) smooth and (b) rough surface

The flow over a flat plate at zero incidence starts as laminar and transitions into the turbulent regime after a certain distance from the plate's leading edge. It can be considered fully turbulent for a Reynolds number of 5×10^5 [1, p. 935]. The TBL grows faster than a laminar boundary layer (equation 4), reflecting the fact that turbulence diffuses momentum more efficiently than molecular diffusion [1, p. 1052].

$$\delta = \frac{0.37x}{Re_x^{0.2}} \tag{4}$$

The flow velocity is assumed to be zero at the surface, known as the no-slip assumption. Above the surface, the flow velocity increases in the wall-normal direction until it reaches the free stream velocity at height δ. The following three assumptions summarise these boundary conditions, where U_e is the velocity at the edge of the boundary [14, p. 473].

- $u(x,0) = v(x,0) = 0$ (no slip and no through flow at the wall)
- $u(x, y \to \infty) = U_e$ (matching of inner and outer solutions)
- $u\left(x_0, y\right) = u_{\text{in}}\left(y\right)$ (inlet boundary condition at x_0)

Since the TBL has a smooth transition to the outer flow, three thickness definitions are specified to describe the boundary layer's character [14, p. 475]. Firstly, δ_{99} is the vertical height where $u = 0.99U_e$ is known as the boundary layer thickness. Secondly, the displacement thickness δ^* (equation 5), which is the height, has the same velocity deficit $U_e - u$ as the actual boundary layer [1, p. 1000-1005].

$$\delta^* = \int_{y=0}^{\infty} \left(1 - \frac{u}{U_e}\right) dy \tag{5}$$

The velocity defect $U_e - u$ is a law used to understand how the velocity inside the boundary layer changes as a function of the distance normal to the wall and the free stream velocity [14, p. 651-653].

Thirdly, the momentum thickness θ (equation 6) is where $\rho U^2 \theta$ is the momentum loss in the actual flow because of the presence of the boundary layer [1, p. 1000-1005].

$$\theta = \int_{y=0}^{\infty} \frac{u}{U_e} \left(1 - \frac{u}{U_e}\right) dy \tag{6}$$

These boundary layer thicknesses are used for systematic fitting across different experiments with physically similar conditions by introducing the following dimensionless Reynolds numbers: $Re_\tau = (\delta U_e / \nu)$, $Re_\delta = (\delta^* U_e / \nu)$ and $Re_\theta = (\theta U_e / \nu)$.

2.2 Features of viscous flows

Theoretical investigations of fluid mechanics primarily focus on potential flow, a concept that considers an ideal fluid, i.e., an inviscid and incompressible fluid [15, p. 3]. Potential flow depicts a lack of tangential forces (shear stresses) between adjacent layers and focuses only on normal forces (pressure). D'Alembert's paradox predicts that a body moving steadily through an infinitely extended incompressible fluid at rest will experience no drag [1, p. 255-260]. Such a result is unacceptable when considering a real fluid[1], which experiences a shear force because of a physical property called the viscosity of the fluid [17, p. 50].

The nature of viscosity causes fluid particles to interact with each other and adjacent surfaces. As a fluid comes in contact with a surface, it adheres to that surface and moves at the same speed; this assumption is the no-slip condition. A general expression for the stresses that occur in fluid flow is given by the Cauchy stress tensor ($\mathbf{t} = \sigma_{i,j} n_j$), where shear stress τ acting on this fluid layer is then expressed by Newton's law of friction (7) [18, p. 36-38].

$$\tau_w = \mu \left(\frac{\partial u}{\partial y}\right)\bigg|_{y=0} \tag{7}$$

[1]Air can be regarded as both ideal and real fluid, depending on the context. When investigating aerodynamic drag and turbulence, the air is considered as a real fluid [16].

Here, the velocity gradient ($\partial u/\partial y$) is the velocity normal to the surface, and μ is the viscosity. An alternative viscosity expression is $\nu = \mu/\rho$, which is the kinematic viscosity. This paper uses the viscosity of air with a value of 1.789×10^{-5} [Kg/ms] and sea level conditions.

2.3 Wall-bounded turbulent flows

The TBL flow that develops from nominally uniform flow over an object is considered wall-bounded. The study of turbulence is usually restricted to relatively simple model flows before applying that information to more complex flows [13, p. 209]. In this case, analysing the TBL on a flat plate at zero incidence (simple flow) will provide valuable insight into the effect of turbulence on a curved surface (complex flow).

2.3.1 Stress behaviour at walls

At the wall, the velocities in the momentum equation are zero based on the no-slip assumption; thus, pressure gradient and stress gradient terms balance (equation 8) [13, p. 332].

$$\frac{\partial P_w}{\partial x} = \frac{\partial}{\partial y}\left(\mu\frac{\partial U}{\partial y}\right)\Bigg|_{y=0} \tag{8}$$

The balance is based on an order-of-magnitude argument, suggesting that the pressure gradient term is in the same order as the convective terms (equations 3). This assumption becomes less valid as the wall-normal distance increases and the stress term becomes significant (equation 9).

$$\frac{\partial P_w}{\partial x} = \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial U}{\partial y}\right) - \rho\overline{u'u'}\right]$$

Since at $y=0$, the term u', $v'=0$ and $\mu\frac{\partial U}{\partial y} = \tau_w$.

$$\mu\frac{\partial U}{\partial y} - \rho\overline{u'u'} = \tau_w \tag{9}$$

Flow behaviour changes with wall-normal distance are known as scaling laws derived from Von Karman's approximate momentum integral approach that is used to analyse viscous boundary layers [13, p. 325-330]. It applies to zero pressure gradient (ZPG) cases like a flat plate at zero-incident, where the flow parallel to the surface experiences uniform flow. Or a curved surface, which may exhibit an adverse pressure gradient (APG) to satisfy the Navier-Stokes [13, p. 325-330].

In terms of stress, viscous and turbulent stresses create a constant-stress region near the wall for a ZPG, whereas a APG scales differently with wall-normal distance (Figure 3).

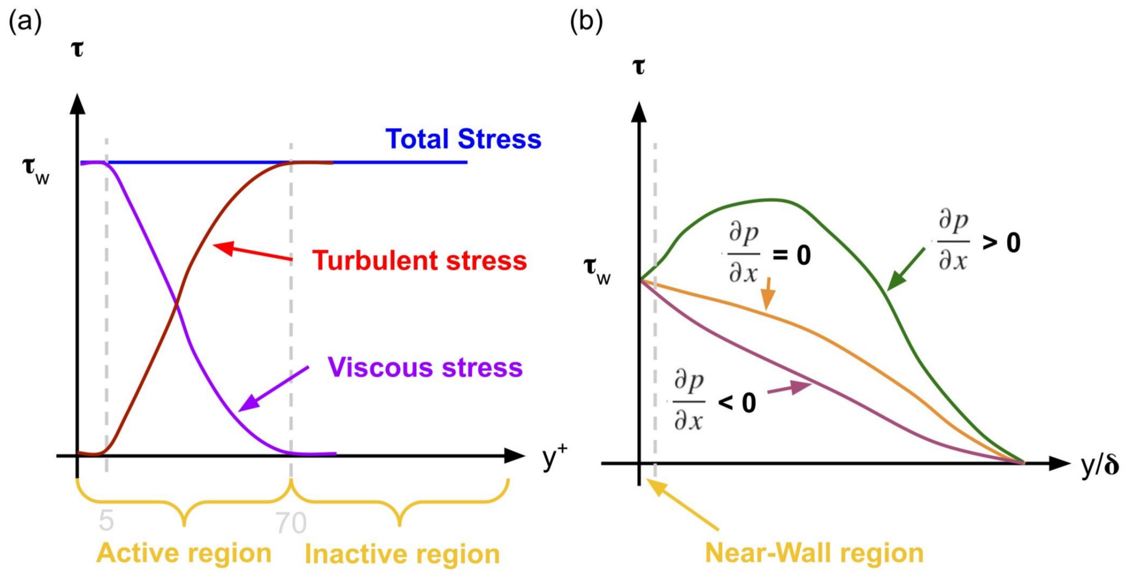


FIGURE 3: Total shear stress distributions (a) ZPG in near wall region (b) In full TBL

Near the wall, the flow is dominated by viscous forces; however, as the distance increases, the impact of turbulent stress ($-\rho\overline{u_i' u_j'}$) on momentum transport becomes more dominant. The fluctuating terms in the turbulent stress term cause a lack of understanding of the flow's general behaviour, causing the process of quantifying stress for an arbitrary pressure gradient to be quite challenging [13, p. 333].

2.3.2 Mean flow over smooth wall

Despite the chaotic nature of turbulence, the majority of the flow can be described by mean flow characteristics [2, p. 5]. The bulk properties of the mean velocity in both smooth-wall and rough-wall boundary layers are derivable by a classical asymptotic matching [2] process [19, p. 2]. Let U_τ denote the friction velocity ($\sqrt{\tau_w/\rho}$) where τ_w is the shear stress at the wall. Suppose also that the flow is an equilibrium state [20, p. 505 - 540], in which δ and U_τ vary sufficiently slowly in the x-direction such that they can be considered as local scales at any particular distance x. Matching

[2] An approach to finding an approximate solution to an equation, or system of equations

these two parameters asymptotically suggests the existence of the overlapping regions mentioned earlier, outer and inner layer, along with a set of length scales characterizing the surface [2, p. 5].

The presence of the wall alters the regime of the turbulent flow near the wall and causes the formation of several distinct layers within the boundary layer [14, p. 648 - 654]. The inner layer consists of three sublayers: the viscous sublayer, the buffer layer, and the logarithmic layer (Figure 2). The viscosity effects are always present near the wall where turbulent fluctuations go to zero, making it the active flow region [14, p. 648]. On the other hand, the outer layer is an inactive turbulent region influenced by the wall geometry [2, p. 5]. The mean velocity profile (Figure 4) spanning these two regions of the TBL is described in terms of the dimensionless velocity U^+ and logarithmic scale of the wall-normal distance y^+. The graph roughly showcases the observed trend based on direct numerical simulations (DNS) from [21] and experiment results from [22] for similar Reynolds numbers.

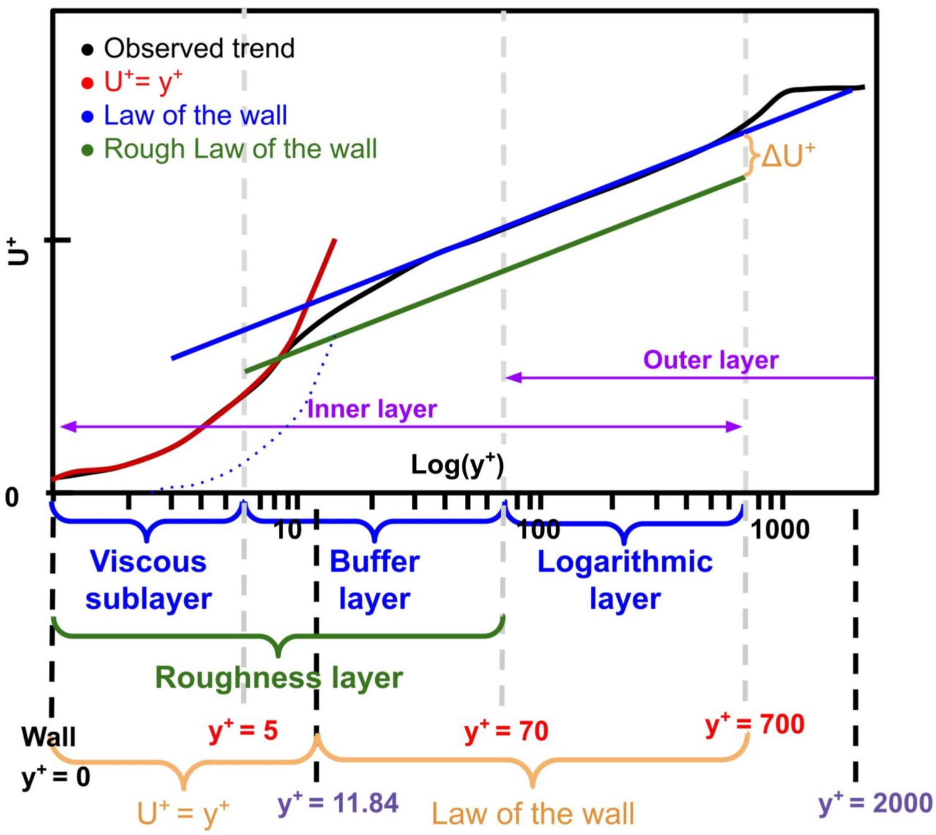


FIGURE 4: Mean velocity profile for adverse and ZPG boundary layer flow

Physical considerations imply that the velocity profile near the wall depends only on the flow parameters and not U_∞. This means viscous forces dominate the wall-normal velocity fluctuations, and the flow is attached to the surface. Due to the high

Reynolds number, the sublayer is also thin enough to suggest that the stress is uniform within the layer and equal to the wall shear stress τ_w (equation 10) [14, p. 651].

$$\mu\left(\frac{\partial U}{\partial y}\right) = \tau_w \rightarrow U = \frac{y\tau_w}{\mu} \text{ or } U^+ = y^+ \tag{10}$$

The relation can be seen as linear and holds approximately until a value of $y^+ = 5$ [15, p. 525]. On the other hand, the overlap of the inner and outer layers causes the mean velocity profiles in the logarithmic layer to scale differently (equation 11). The velocity distribution in this region, valid for the range $70 \leq y^+ \leq 700$ [15, p. 525], is termed as the "law of the wall" [3] [14, p. 652].

$$U^+ = \frac{1}{\kappa}\ln(y^+) + C \tag{11}$$

Here, κ is the *von Karman* constant ($k \approx 0.41$), and the value of constant C is approximately 5.1 [2, p. 5]. The turbulence fluctuations greatly exceed the viscous forces in this region.

Lastly, in the region of $5 \leq y^+ \leq 70$ known as the buffer region, both viscous and turbulent forces are significant, causing the velocity profile to be neither linear nor logarithmic [14, p. 653]. This region can be modelled using the two standard wall functions mentioned above, intersecting at $y^+ \approx 11.84$ [23]. This approach is not exact, prompting other wall functions. One that models the whole inner layer is called the Spalding/Reichardt law [2, p. 6]. Another wall function that aims to model the outer layer in mode detail is called the law of the wake [24]. These laws are of no particular interest to this study and have not been explored.

2.3.3 Turbulent flow over rough surfaces

Rough surfaces, unlike smooth ones, impune the formation of the viscous sublayers due to the increased average height of the surface [2, p. 11]. An extreme example is the atmospheric boundary layer, where buildings and other obstacles lead to wake formation behind each element [14, p. 660]. Despite many studies on this topic, a comprehensive understanding of such flows remains elusive, and the flow must be characterised using empirical correlations [25, p. 2].

The equivalent sand-grain approach pioneered by Nikuradse [9] attempts to define the roughness using a length scale of spheres (Figure 5). The height of a uniformly rough surface is described by the roughness height k. However, a single parameter cannot accurately describe the surface when the roughness shape significantly differs from the sphere-shaped sand grain. For this reason, the equivalent

[3] Also referred to as "Log law"

sand-grain roughness k_s introduced by Colebrook and White [26, p. 373] is used.

For this study, the simplification that the tyre's surface consists of a uniform and sphere-like topology is made. Therefore, using the sand-grain approach does not represent the exact case. More sophisticated empirical correlations for other surface geometries can be found in [2, p. 14]; however, attempting to apply these to the surfaces at hand is outside the capabilities of this study.

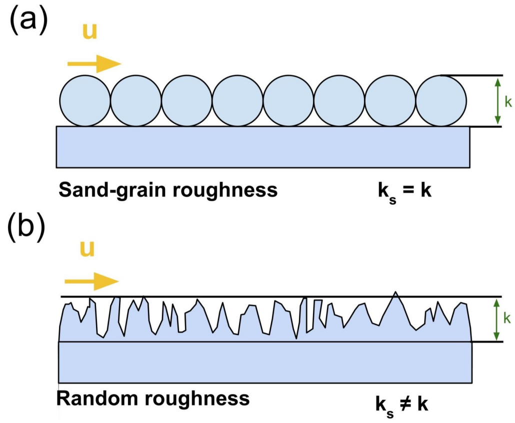


FIGURE 5: Equivalent sand-grain (a) uniformly distributed (b) randomly distributed surface

2.3.4 Roughness regimes

The roughness of a surface impacts the extent to which viscosity affects the flow. Hence, the inner layer is now considered as two sublayers: the roughness sublayer and the logarithmic sublayer (Figure 2). The logarithmic sublayer remains the same as in the smooth case, however the roughness sublayer combines the buffer and viscous sublayer to account for the sand grain roughness. This is achieved with the use of roughness Reynolds number k_s^+ shown in equation 12 [2, p. 12].

$$k_s^+ = \frac{k_s U_\tau}{\nu} \tag{12}$$

Based on the influence of the surface's roughness, k_s^+ depicts the extent to which the roughness influences the viscous sublayer using three regimes. Firstly, smooth $k_s^+ \leq k^+_{\text{smooth}}$ where the surface can be considered smooth ($k_s = 0$). Secondly, transitional regime $k^+_{\text{smooth}} \leq k_s^+ \leq k^+_{\text{rough}}$ where both viscosity and roughness affect the skin friction, and lastly, fully rough $k_s^+ \geq k^+_{\text{rough}}$. The two critical limits of the roughness Reynolds number k^+_{smooth} and k^+_{rough}, which define the following three regimes, can be found for different roughness types and shapes in literature. For the purpose of this study, the original values proposed by Nikuradse [9] of 5 and 70 are utilised (Figure 4).

2.3.5 Mean flow over rough walls

Current engineering methods for modelling rough surface turbulence build upon the described mean flow over smooth walls. A change proposed by Clauser [27] is an adjustment to the law of the wall, as shown in equation 13 [14, p. 660].

$$U^+ = \frac{1}{\kappa}\ln(y^+) + C - \Delta U^+ \tag{13}$$

Modifying the law of the wall provides a roughness correction in the form of a downward displacement, as seen in Figure 4. The new term ΔU^+ was originally defined by Hama [28], assuming a ZPG flow. Even though Perry & Joubert [29] have shown that the function may be used in boundary layer flows with APG, the validity and accuracy can be compromised. The function is expressed in equation 14, where A and B are constants [2, p. 16].

$$\Delta U^+ = \frac{1}{A}\ln(k_s^+) + B \tag{14}$$

The function also varies depending on the roughness regime k_s^+ of the flow, hence why the roughness function ANSYS Fluent [23] will provide this relation when conducting numerical simulations in upcoming section 3.

3 Numerical Simulation

CFD is an essential tool in engineering used to simulate flow alongside experiments and theoretical analysis to support the research [3, p. 1]. Numerical turbulent flow analysis is commonly accomplished using one of the following three established CFD methods.

- **direct numerical simulations (DNS):** Resolves the complete flow, making it very computationally heavy. Often available from databases [3, p. 207-233].
- **large eddy simulations (LES):** Concentrates on resolving large scales of turbulence directly and modelling the remainder [3, p. 269-306].
- **Reynolds-averaged Navier Stokes (RANS)**: Models the entire flow by time averaging and evaluating the effect of fluctuations by turbulence modelling [3, p. 237-267].

This study has chosen RANS primarily for its efficiency and practicality [3, p. 237]. RANS has the lowest computational cost of the three options, making it suitable for steady flows that must be computed for several roughness cases. However, RANS's weakness is its lower accuracy and sensitivity to modelling choices, such as the cell resolution [30, p. 122-125].

3.1 Methodology

The commercially available CFD software ANSYS Fluent computes the RANS equations shown in section 2 for the flow case considered. Additional equations or assumptions introduced by picking a turbulence model are necessary to solve the previously mentioned turbulence closure problem. This study utilises the shear stress transport (SST) $k-\omega$ model from the eddy viscosity class of turbulence models for this purpose. This model is the successor to the $k-\omega$ and $k-\epsilon$ models [31, p. 1602] and has been chosen because studies indicate that it shows better agreement with experiments in external flow when predicting turbulence characteristics near the wall [32, p. 108]. Numerical studies with RANS often compare different turbulence models as a form of validation; however, this simulation aims to replicate a tyre's geometry, which is challenging to model accurately. In the upcoming comparison between the simulation and wind tunnel analysis, these inaccuracies are expected to exceed any differences in turbulence in the models. Therefore, this study does not compare different turbulence models.

The Semi-Implicit Method for Pressure-Linked Equations (SIMPLE) algorithm is the solver of choice for the steady incompressible flow, which is the case here. SIMPLE is an iterative method that solves for the three unknown directions of the velocity field U_x,U_y,U_z and mean pressure P using discretised RANS [3, p. 78-79].

The finite-volume method discretises the spatial terms of the RANS equations by integrating over the control volume (CV) to produce discrete cells. The centre of a cell called the centroid, stores flow quantities such as velocities and pressure when computing the solution. The second-order upwind scheme then discretises the non-linear convection term in the momentum equations [30, p. 129]. Upwind differencing is a popular choice for convection-dominated flows, as the case here, because it provides a stable solution based on the direction of the mass flux [33, p. 253].

The stability and convergence of the CFD simulation are ensured by setting the Courant-Friedrichs-Lewy (CFL) condition to 0.9. This adaptive scheme adjusts the time step based on the local flow velocity and grid size to uphold the CFL condition across the CV [3, p. 55-65].

3.1.1 Simulation setup

The computational domain (Figure 6) captures the flow case using a CV approach. The fluid domain has a total length of 100 [cm] and is positioned above the tyre of diameter D.

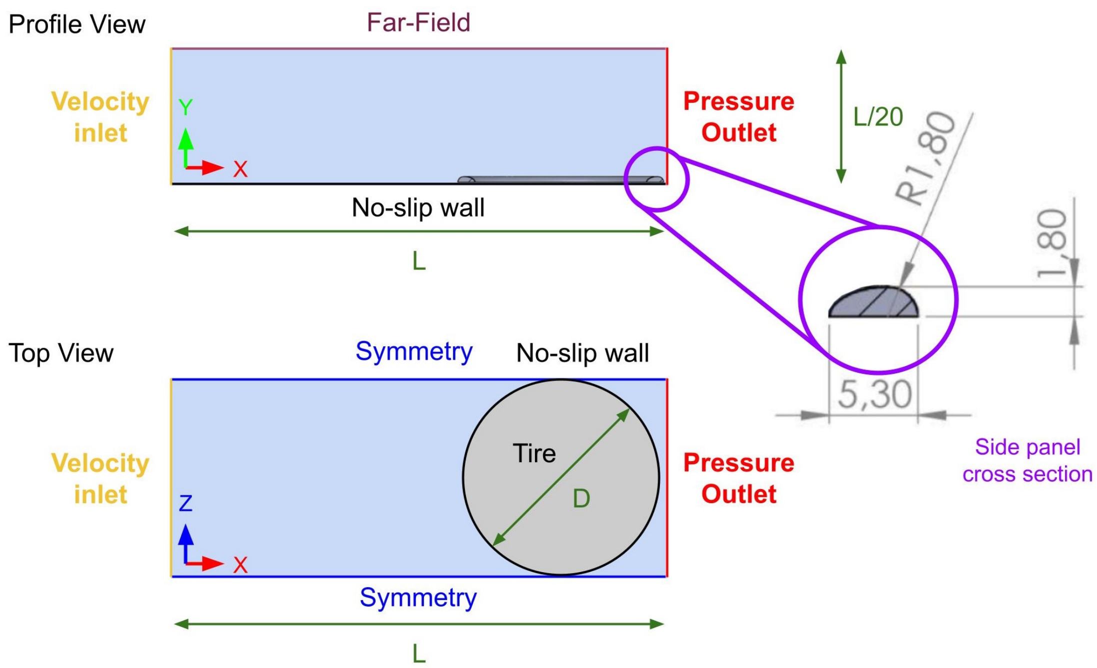


FIGURE 6: Computational domain for CFD simulation with boundary conditions and dimensions

The airflow enters the CV with a speed of 65 [km/h] (≈ 17.7) [m/s] at the velocity inlet, where the boundary condition is set to the free stream velocity $\left(u(x=0, y, z) = U_\infty\right)$. It then continues through the CV until the outlet, which specifies that the ambient pressure is reached $\left(\frac{\partial u_i}{\partial x_i}\Big|_{x=L} = 0\right)$. The flow is considered symmetrical on either side of the CV along with a far-field boundary condition on the top stating that the freestream velocity is reached $(u(x, y = y_{\max}, z) = U_\infty)$. Lastly, no-slip boundary conditions $(u(x, y = 0, z) = 0)$ are also applied to the tyre and bottom, modelled as separate surfaces [23]. The tyre's side panel cross-section (Figure 6) is removed from the CV to constrain the fluid.

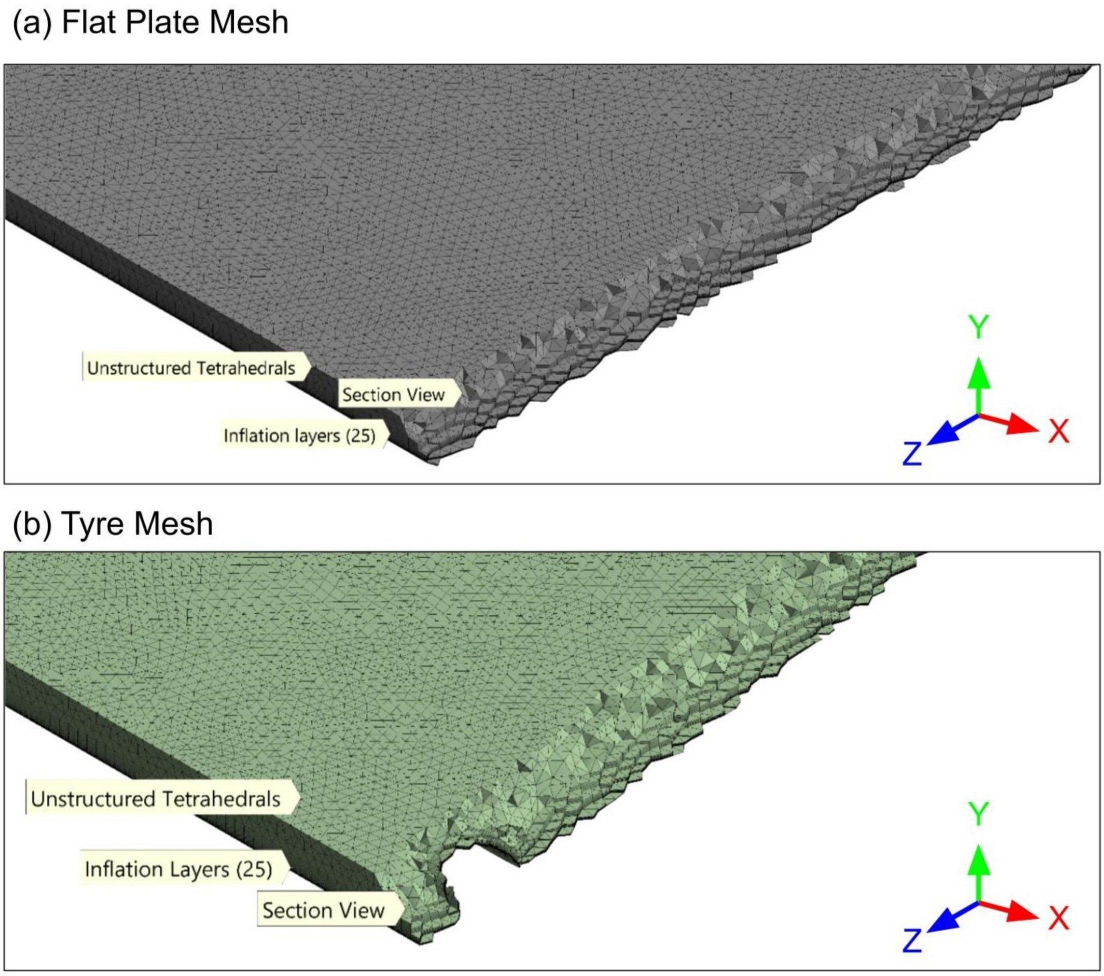


FIGURE 7: Mesh of CV for (a) flat plate and (b) tyre

A process called meshing, achieved using the built-in tool of ANSYS meshing, accomplishes discretising the CV into cells of unstructured tetrahedrals. Inflation layer meshing is implemented near the wall to resolve the velocity profile in the inner sublayer of the TBL. These layers of cells progressively decrease in size, allowing for better capture of the steep velocity gradient present there. The first of twenty-five inflation layers is set at $\Delta S = 0.02$ [mm] to obtain $y^+ < 1$. This height is determined by equation 15, where U_τ is determined using flat-plate boundary layer theory $Re_x = \frac{\rho U_\infty L}{\mu}$ and $C_f = \frac{0.074}{Re_x^{1/5}}$ from [1, p. 1051-1062].

$$\Delta S = \frac{y^+ \mu}{\rho U_\tau} \tag{15}$$

3.1.2 Estimating wall shear stress

Even though ANSYS Fluent can calculate the wall shear stress τ_w needed to determine U_τ, predicting such values causes problems with accuracy [1, p. 1072-1077]. Some reports indicate that the errors in estimating U_τ and ΔU^+ are 3%–5% and 6%–10% respectively [34, p. 335]. If more accuracy is needed, the Clauser method can be implemented, indirectly estimating the local mean skin friction in a TBL based on velocity defect profile [35]. In this study, the numerical simulations are used to support the findings in the wind tunnel analysis, and a larger degree of accuracy is not necessary. Therefore, the wall shear stress values from ANSYS Fluent are considered sufficient.

3.1.3 Simulation cases

Two types of surface cases, the smooth wall case and the rough wall case, are investigated for both flat plate and tyre geometry. Four different tyre roughnesses Figure 8 are available for the wind tunnel test described in section 4.

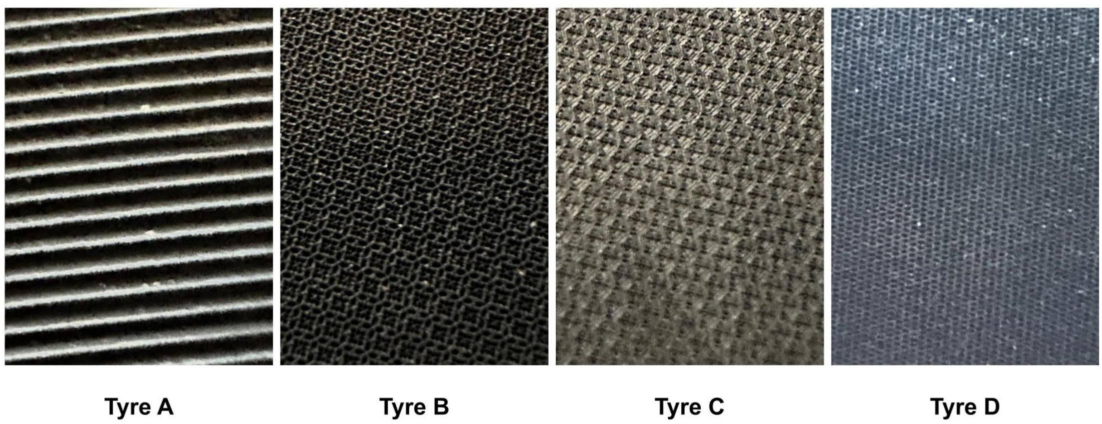


FIGURE 8: Close up of tyre's side panel surfaces for four different tyres

The roughness of the different surfaces is captured in the simulation using the previously described sand grain approach for uniformly rough surfaces. A vernier calliper has been used to measure the height k_s of the different surfaces to obtain the roughness height values (Table 1).

Geometry	Surface	k_s [mm]	k_s^+
Flat Plate / Tyre	Smooth	0	0
	Tyre A	1	47.11
	Tyre B	0.5	23.55
	Tyre C	0.25	11.78
	Tyre D	0.05	2.36

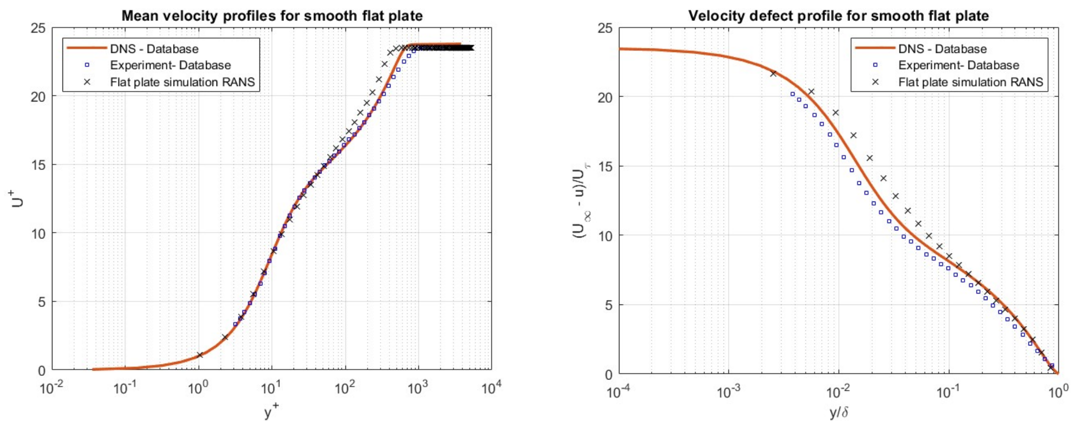
TABLE 1: Table showing simulation cases for flow case

3.2 Numerical results

The flat plate geometry validates the model by looking at the flow data at the trailing edge. Meanwhile, the results from the tyre geometry determine the skin friction drag exerted on the tyre's side panel.

3.2.1 Flat plate validation

A comparison of the smooth wall numerical simulation with DNS data from [21] and wind tunnel data from [22] review the validity of the CFD results acquired in this study. The mean velocity profile (Figure 9a) of the simulation shows great agreement with the databases by collapsing onto the observed data. The velocity defect profile (Figure 9b) differs slightly from the DNS data, but resembles the expected trend.



(A) (B)

FIGURE 9: Comparison between DNS data from [21], experimental wind tunnel data from [22] and RANS simulation for a smooth wall flat plate at trailing edge with similar Reynolds numbers. (A) Mean velocity profile (B) Velocity defect profile.

TBL thicknesses (equations 5, 6) are computed from the simulated velocity profiles (table 2). For example, the simulation's velocity profile for the smooth-walled flat plate at the trailing edge (Figure 10) determines the thicknesses by calculating the difference in data points from the wall-normal dy and numerically integrating the equations.

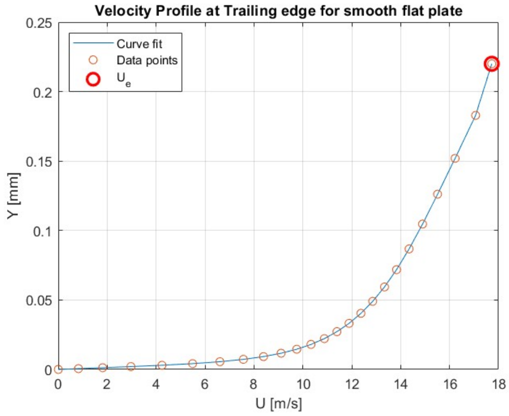
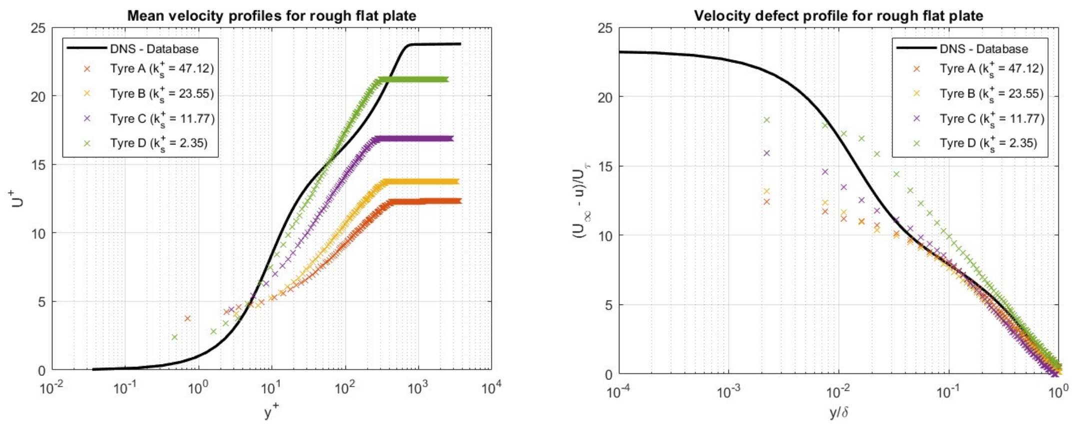


FIGURE 10: Velocity profile for smooth flat plate at trailing edge

Mean velocity and defect profiles (Figure 11) for the rough flat plate cases illustrate the changes with surface roughness. The plots indicate that the mean velocity profiles experience an increase in the Hama roughness function ΔU^+ shown by the vertical shift downwards.

k_s^+	U_τ [m/s]	δ_{99} [mm]	δ^* [mm]	θ [mm]	Re_τ	ΔU^+	C_f	ΔC_f
0	0.765	2.21	0.048	0.012	2342	0	0.00374	
47.12	1.038	4.85	0.648	0.239	5877	5.21	0.00688	0.00314
23.56	0.956	4.45	0.635	0.230	5392	3.54	0.00584	0.00104
11.77	0.818	3.93	0.591	0.208	4786	2.56	0.00427	0.00156
2.35	0.688	3.59	0.558	0.198	4165	1.40	0.00302	0.00125

TABLE 2: CFD simulation results for flat plate at trailing edge



(A)

(B)

FIGURE 11: Rough wall cases of flat plate for different surface roughnesses, with DNS data from [21]. (A) Mean velocity profile (B) Velocity defect profile.

The roughest surface, tyre A, is seen to experience the most friction drag with

$\Delta U^+ = 5.21$. The difference is doubled with a coefficient of 0.00688 compared to the smoothest surface of 0.00314. However, the roughness of tyre D's surface is seen to experience less friction drag than the smooth surface, even though $\Delta U^+ = 1.40$. This is most likely due to the previously mentioned error in estimating the skin friction drag using ANSYS because a leftward shift is seen.

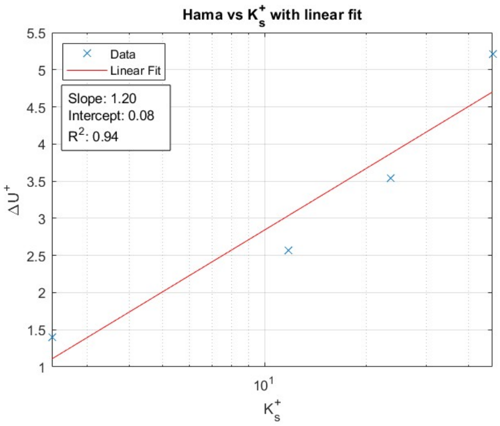


FIGURE 12: Hama roughness function ΔU^+ as a function of roughness number k_s^+

Plotting the Hama function ΔU^+ is against the logarithmic scale of the roughness number k_s^+ (Figure 12) illustrates the effects in roughnesses (equation 14). A line of best fit is plotted to determine the slope and intercept corresponding to constants A and B (equation 14). These are determined to be 1.20 and 0.08 respectively. A coefficient of determination R^2 of 0.94 also can be seen in the trend fitting, indicating a good fit. With this, the model follows the expected behaviour and is validated.

3.2.2 Simulation results

The behaviour of the flow for the tyre geometry is analysed by inspecting the velocity profile in the two-dimensional cross-section of the CV at the centre of the tyre ($z =$ 31.5 [cm]) (Figure 13).

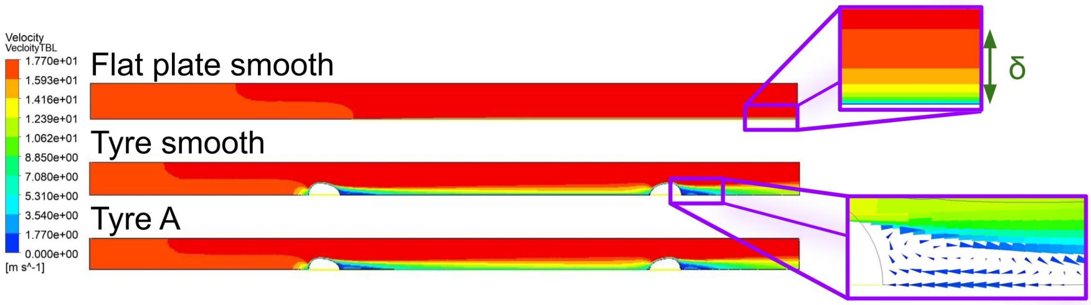


FIGURE 13: 2D velocity profile showing TBL for (a) smooth flat plate (b) Smooth Tyre (c) Tyre A, at direction $z = 31.5$ [cm]

Upon closer inspection of the flow after the tyre's side panel, the flow is in reverse direction. Therefore, a APG is present, causing the flow to separate and introduce pressure drag by forming a wake[4]. Further detail is shown in the pressure distribution along the tyre's profile (Figure 14), illustrating the pressure gradient's implications on the flow behaviour. The coefficient of pressure C_p (equation 16) is used for this purpose. If $C_p \approx 0$, the local pressure is approximately equal to the freestream pressure P_∞, whereas a value $C_p < 0$ depicts flow separation and recirculation zones since pressure gradient is negative [1, p. 369-390].

$$C_p = \frac{P - P_\infty}{\frac{1}{2}\rho U_\infty^2} \tag{16}$$

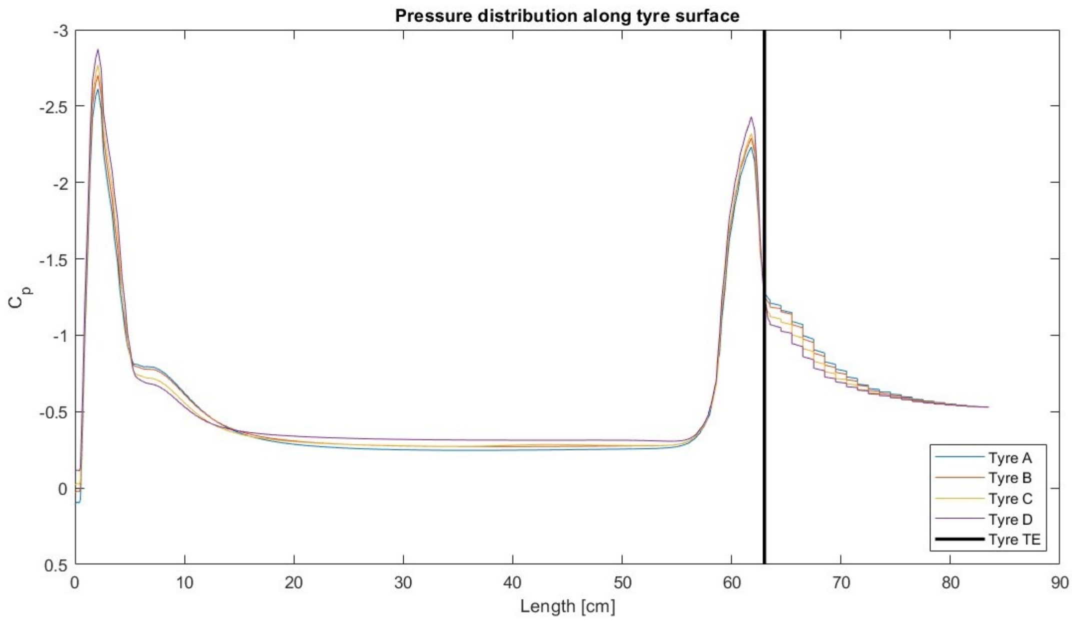


FIGURE 14: Coefficient of pressure along tyre's surface at direction $z = 31.5$ [cm]

CFD results for the tyre geometry include the skin friction and pressure drag taken directly from ANSYS (Table 2).

Geometry	Surface	k_s^+	Skin friction [N]	Pressure drag [N]	Total drag [N]
Flat Plate	Smooth	0	0.305	0	0.305
Tyre	Smooth	0	0.230	2.393	2.623
	Tyre A	47.11	0.617	3.107	3.724
	Tyre B	23.55	0.497	2.885	3.383
	Tyre C	11.78	0.369	2.708	3.077
	Tyre D	2.36	0.237	2.383	2.620

[4] Region of eddying fluid flow that forms downstream of an object after flow separates from its surface

As the theoretical analysis suggests, the results indicate that skin friction increases with surface roughness. The difference in skin friction between the roughest and smoothest tyre is 0.380 [N]; however, as discussed in section 3.2.1, the wall shear stress for tyre D may be flawed. Furthermore, the simulation results indicate a decrease in friction drag for the two geometries with a smooth surface. This is likely due to the flow separation on the tyre, which causes a lower contribution of friction drag since there is less surface area in direct contact with the flowing fluid, as explained in section 2.

Pressure drag is also seen to increase with surface roughness. Rougher surfaces disturb the airflow, causing a premature separation that creates a larger wake region and pressure drag. The impact of pressure drag is greater than fiction drag and differs by 0.724 [N] for two extreme roughnesses.

4 Wind Tunnel Analysis

Wall-flow experimental designs for measuring shear stress can be categorised into indirect and direct methods [36, p. 875-886]. Indirect methods extract wall shear stress from other flow properties related to shear stress, such as pressure or temperature. For instance, a preston tube obtains the wall shear stress by measuring the pressure distribution [37, p. 173-174]. The disadvantage of this method is the necessity to modify the wall object, hence unwillingly influencing the flow behaviour. hot-wire anemometry (HWA) and laser doppler velocimetry (LDV) and are examples of less intrusive indirect methods which calculate the wall shear stress from measured flow field velocities [37, p. 136-185].

Conversely, direct methods accurately measure wall shear stress in complex, difficult-to-model flows because no assumptions about the fluid properties, flow field or surface need to be made [36, p. 883]. In theory, the simplest device is a floating element (FE) sensor that estimates the skin friction drag directly by measuring the displacement of the object caused by the airflow [36, p. 875-886]. This section describes designing a FE set-up to measure the skin friction drag of the tyre's side panel.

4.1 Methodology

In essence, FE sensor measures the streamwise net force F_w that acts on the tyre's area S. The tyre is structurally independent of the test section, meaning it is free to move in the direction of the flow because of an air gap surrounding it. Placing a load cell such that the displacement is gauged and converted into an electrical signal allows the net force to be measured. The mean wall shear stress is then $\tau_w = F_w/S$.

4.1.1 Design considerations

Even though a FE sensor is a simple device in theory, there are several practical challenges, making it difficult to implement [38, p. 53]. The following challenges are significant error sources, predominantly of systematic nature, which the design aims to address [39, p. 155].

(i) The tyre must have a large enough surface area to exert a net force of significant magnitude that can be measured accurately, yet small enough to measure local conditions [36, p. 883].

(ii) Unwanted contributions from the air gap around the FE causing static pressure differences on non-parallel surfaces.

(iii) Uncertainty caused by misalignment of the FE with the sensor.

(iv) Errors associated with pressure-gradient forces caused by complex geometry, which add pressure drag and forces in undesirable directions.

For obvious reasons, the tyre's surface area can not be altered, and instead, the design addresses (i) by selecting a load cell capable of measuring the expected load from numerical solutions (0.5 - 4 [N]). For this purpose, the loadcell type CB17-1K-11 [40] with a rated capacity of 9.801 [N] (1kgf) was selected.

To address (ii), studies suggest approximating the net force components in directions aside from the flow, caused the flow escaping throguht the air gap [36, p. 875-886]. Alternitely filling the gap with a compressible material [39, p. 155] to reduce the flow of air leaving the set-up. Instead, this design implemented a chamfered edge on the testing surface to reduce the amount of airflow escaping [41, p. 59], and any remaining influence was considered negligible based on the tyre's weight constraining the tyre's movement.

The design minimises misalignment errors (iii) by using four flexures that constrain the movement to one degree of freedom (DOF), the flow direction. This design choice does not fully eliminate alignment error, but by overconstraining the system, the assumption is made that it is not of great significance.

Flow over a curved surface can be expected to develop an APG and lead to unwanted pressure drag measurements (iv). Unfortunately, this cannot be accounted for with this measurement setup. A smoke test was made to check for visual flow separation, however the irregular surface on a tyre's surface induces local pressure variations and small-scale separations that contribute to pressure drag regardless [13, p. 383-401].

4.1.2 Mechanical configuration

The working principle behind the measurement setup (Figure 15) utilized the wind tunnel's test section by suspending the tyre on a FE such that it sat flush with the surface. The FE was constructed to support the weight of the tyres using four aluminium bolcan[5] pieces arranged in a rectangle. The load cell was positioned perpendicular to the FE surface to measure the displacement.

[5] Aluminium beams used in support of wind tunnel [42]

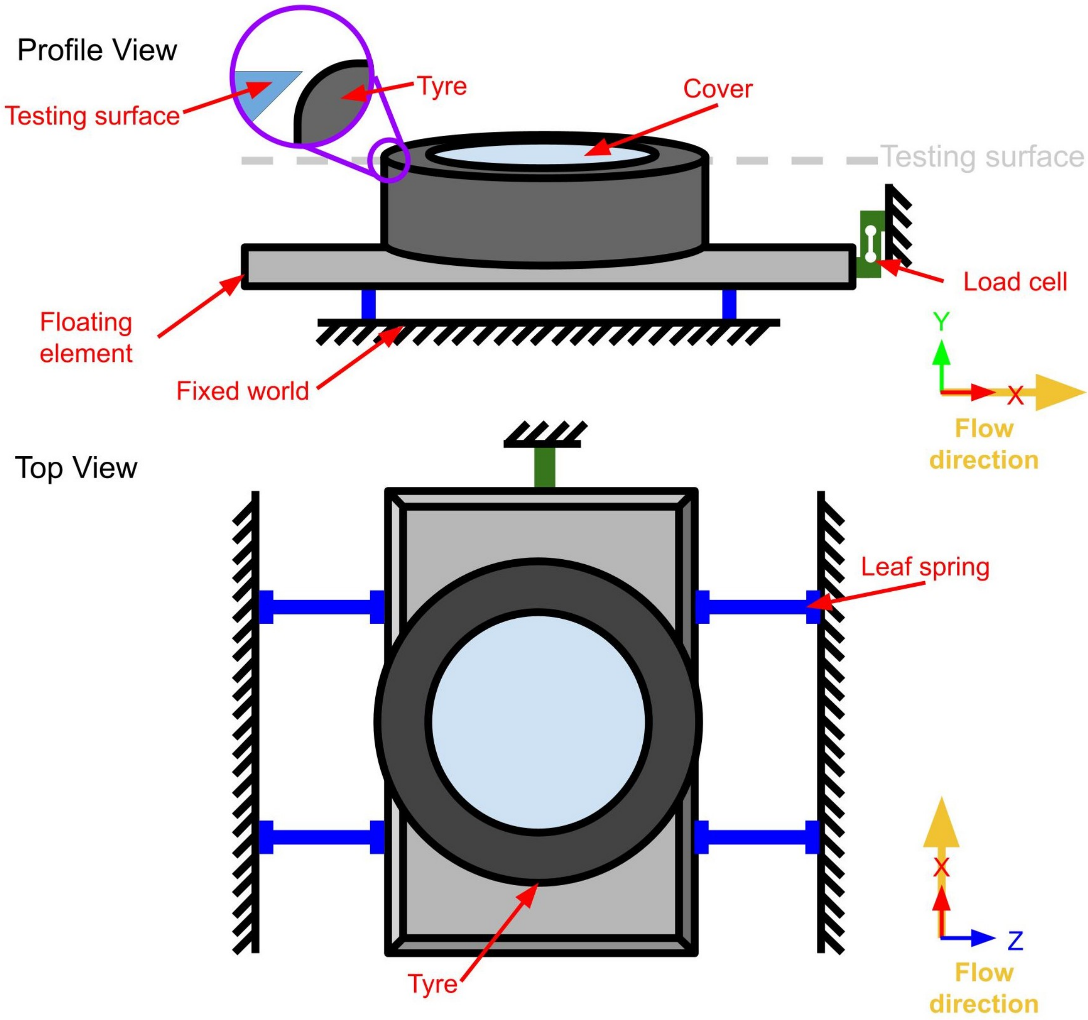


FIGURE 15: Schematic of wind tunnel measurement set-up

Sheet metal with a 0.5 [mm] thickness created the leaf springs that acted as the system's flexures. To attach the FE to a fixed surface and restrict the movement to only one DOF required the leaf springs to possess a low stiffness in the streamwise direction. This is because the stiffness of the flexure k_f is defined as $k_f = R_x/\Delta x$ where R_x and Δx are the reaction force and displacement respectively. Therefore, a lower stiffness yields a larger displacement, which is more easily measured by the load cell, increasing accuracy.

Low stiffness and the tyre's weight potentially causing the leaf springs to buckle were essential factors considered when deciding on the leaf spring dimensions (Table 3, Figure 16). A project member conducted a more in-depth analysis of the leaf springs design and buckling mode in [43].

Dimensions	Length [mm]
L	140
L_S	90
W	0.5
W_S	8
H	50

TABLE 3: Dimensions of four leaf springs used to suspend the FE

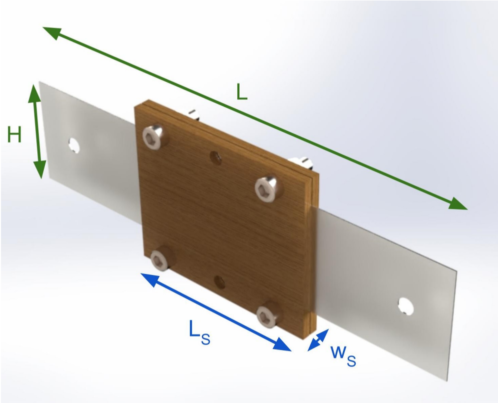


FIGURE 16: Labled Solidworks render of leaf spring element

The testing surface of the wind tunnel was also extended to reduce the air gap around the FE using a plexiglass insert. This insert was made of the same material as the wind tunnel and has a chamfered edge to reduce the flow escaping the test section through the air gap. Any other gaps in the test surface caused by screw threads, sections of the test surface meeting and the cover on the tyre were sealed using masking tape. Covering such cavities with masking tape reduces vibrations and noise that negatively impact the load cell readings [37, p. 579-606].

4.2 Experimental methods

The measurement set-up (Figure 17) was designed to be implemented in the closed-circuit wind tunnel at the University of Twente's Engineering Fluid Dynamics (TFE) research group [44]. This wind tunnel has a 10:1 contraction followed by a 0.7×0.9 [m^2] open test section inside a large anechoic chamber where the experiment was conducted. The freestream turbulence intensity is reported to be lower than 0.08% [45].

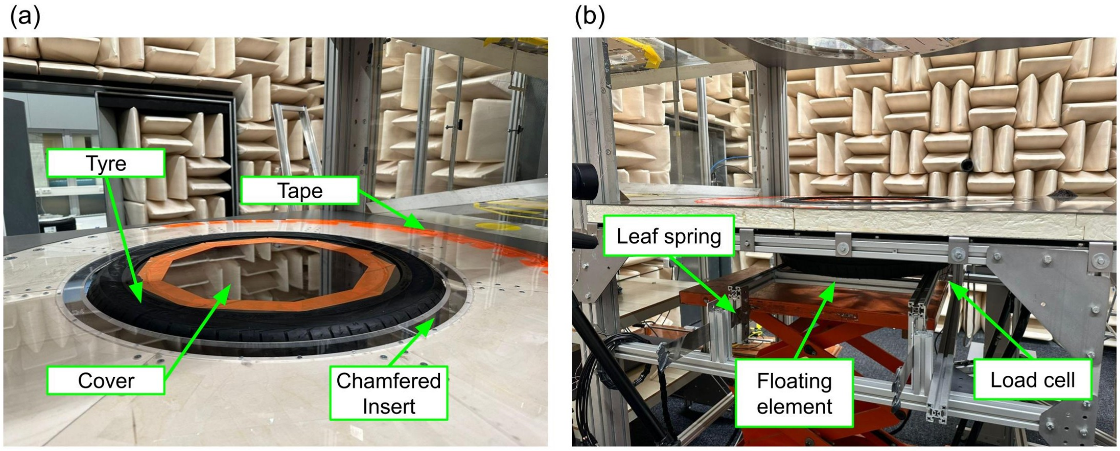


FIGURE 17: Measurment set-up in wind tunnel, (a) top view (b) side view

4.2.1 Data aqusition & uncertainty principles

FE setups can be considered as mass-spring systems with low external damping. The airflow will naturally introduce damping effects, reducing the system's oscillations and gradually reaching a steady state [36, p. 884]. The system's natural frequency f_n is estimated based on the load cell's specified stiffness of 57.69 [kN/m] [40]. Assuming the leaf springs' stiffness is insignificant compared to the load cell and a mass of approximately 30 [kg], the natural frequency is determined to be 6.98 [Hz].

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \tag{17}$$

The interaction between the airflow and the system can lead to complex turbulence behaviours, such as vortex-induced vibrations, which exert periodic forces on the system. If these forcing frequencies are near the eigenfrequency, resonance can occur, leading to large amplitude oscillations [36, p. 884]. To minimize these effects, a large sampling frequency of 50 [Hz] was chosen to capture the detailed dynamics of the system, and the force measurements were averaged once a steady state was reached.

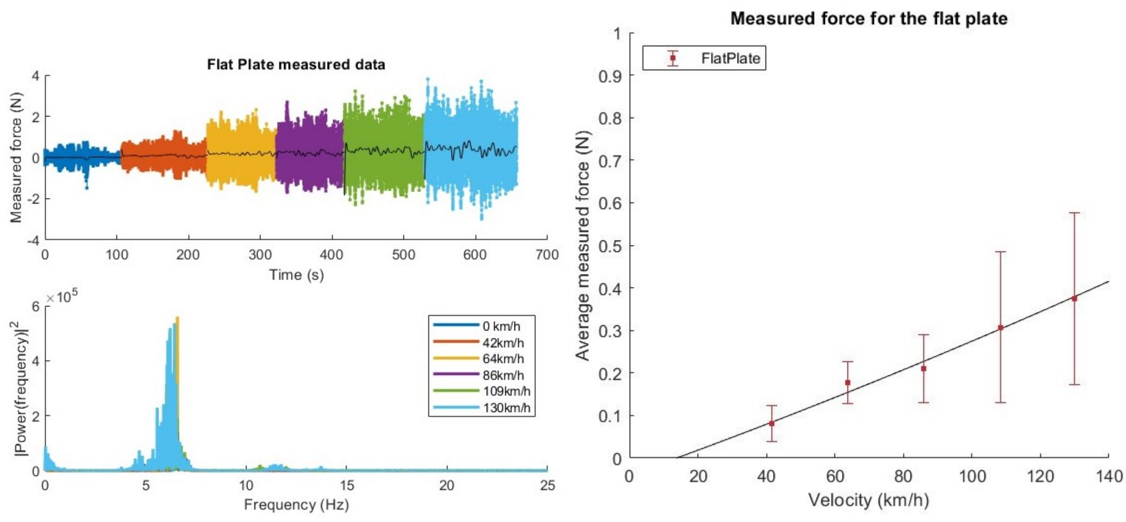
Load cell type CB17-1K-11 [40] was used to record the force. It uses a Wheatstone bridge circuit, calibrated by loading the load cell with a 1 [kg] weight in a static condition. Calibrating, along with pre-tensioning the load cell, allows for pre-amplification, which gives a more desirable high signal to noise ratio (SNR) when converting the measured data from analogue to digital [46, p. 11-103]. A low-pass Butterworth filter was applied to the measured data to improve and attenuate undesirable signal noise further. Such a procedure reduced the chances of the load cell suffering from sensitivity drift and zero offset errors [47].

4.3 Experimental results

The experiment was conducted for six velocities ranging from 0 to 130 [km/h]. Each velocity measurement was recorded for approximately 60 seconds to reach a steady state. The measurements were repeated three times for each of the four tyre roughnesses (Figure 8) and a flat plate. A smoke test was also conducted to provide visual insight into the flow behaviour. Since the experiment set-up suffered from systematic errors, it was conducted on the same day.

4.3.1 Flat plate validation

To validate the experiment, a flat plat was tested to establish a comparison with theoretical expectations. Using the measured data, a low pass filter and the fast Fourier transform were applied to produce a power spectrum (Figure 18a). The power spectrum shows the amplitude of different frequency components and contains a peak at an approximate frequency of 7 [Hz]. These results support the assumption that the leaf springs have a negligible stiffness as the dynamic analysis, which excluded the leaf spring stiffness, expected a natural frequency of 6.98 [Hz].



(A) (B)

FIGURE 18: Measurement results for flat plate experiment (a) Recorded force with time for varying velocities and filtered data using fast Fourier transform showing power spectrum (b) Average measured force for varying velocities

Based on the recorded data, the average measured force increases linearly with velocity (Figure 18b). For a velocity of 64 [km/h], the force reads 0.183 [N], which is lower than the simulated value of 0.305 [N]. Since the error bars don't coincide with the expected value either, it can be said that a systematic error is likely to have occurred, or the measurement set-up may be unreliable.

4.3.2 Exmperiment results

Values obtained using the FE set-up (Figure 19) show an exponential increase in the measured force with velocity. The error bars showcase the sampled data's standard deviation and the load cell's measurement uncertainty.

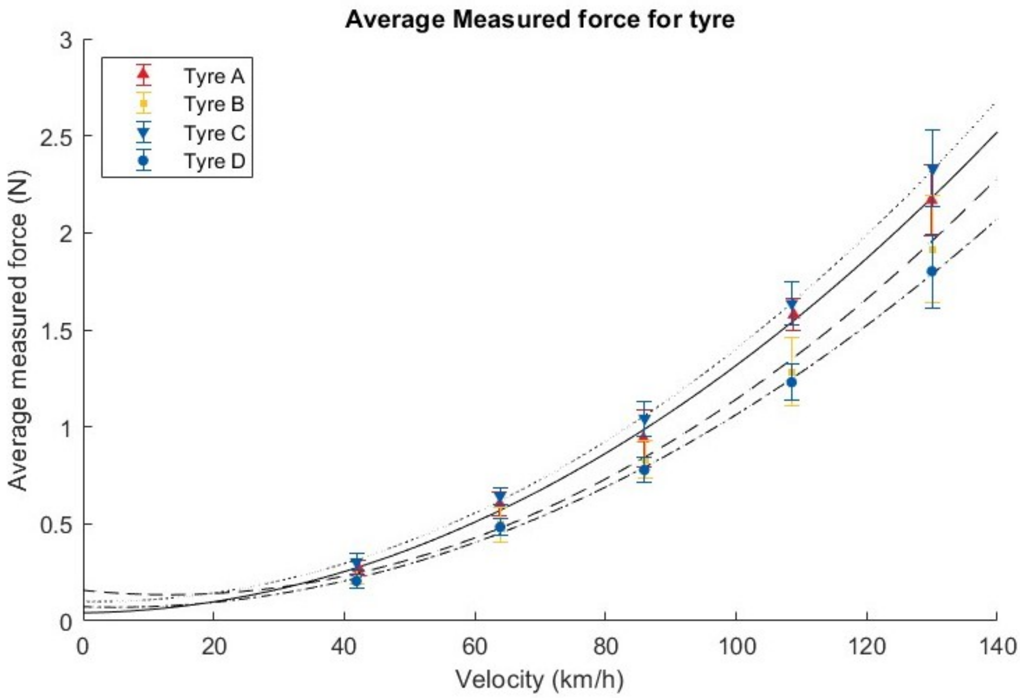


FIGURE 19: Average measured force for four different tyres with varying roughness across a range of velocities

According to the results, tyre C exerts the largest force for all the measured velocities. Aside from tyre C, the measured force increases with surface roughness, as the previously discussed theory predicts. The difference in forces is 0.160 [N]; nonetheless, the error bars overlap, suggesting that measurements could be similar since they are within the uncertainty range. It also implies that the observed differences could be due to random variation rather than a true underlying difference.

FIGURE 20: Smoke test conducted on tyre A at 64 [km/h]

In addition to the measurements, the smoke test (Figure 20) does not show a visible circulating airflow; in principle, this implies no flow separation or pressure drag component in the measured force. However, the smoke test's lack of visible flow separation does not guarantee this. A wake region and local flow separations are almost certain to have occurred based on the irregular surface and curved profile explained in section 2. Despite this, comparing this smoke test to the numerical simulation suggests that the pressure drag maybe be less significant than expected.

5 Results and discussion

The theoretical analysis conducted in section 2 suggests that a rougher surface causes a larger skin friction drag. To verify this expectation, the CFD simulation and wind tunnel analysis are compared by focussing on 64 [km/h] velocity.

5.1 Comparison of methods

The results of the experiment at 64 [km/h] (Figure 21) show the measured force increasing with larger surface roughness. The exception to this expected trend is tyre C, which experiences the largest force of 0.643 [N]. Tyre C has the highest measured force for all the velocities, suggesting that a systematic error, like misalignment of the tyre with the testing surface, may have occurred when mounting the tyre. Although this cannot be stated as certain, such an error would be present for all three measurement trials of tyre C. Therefore, compared with the other tyres, it is improbable that a random error is responsible for this result. Another explanation for the results deviating from the expectation is tyre C's surface may produce more drag.

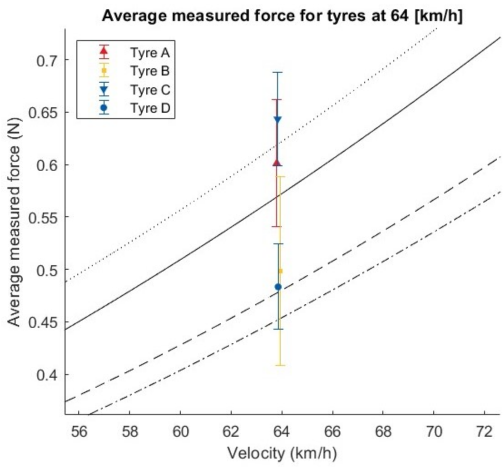
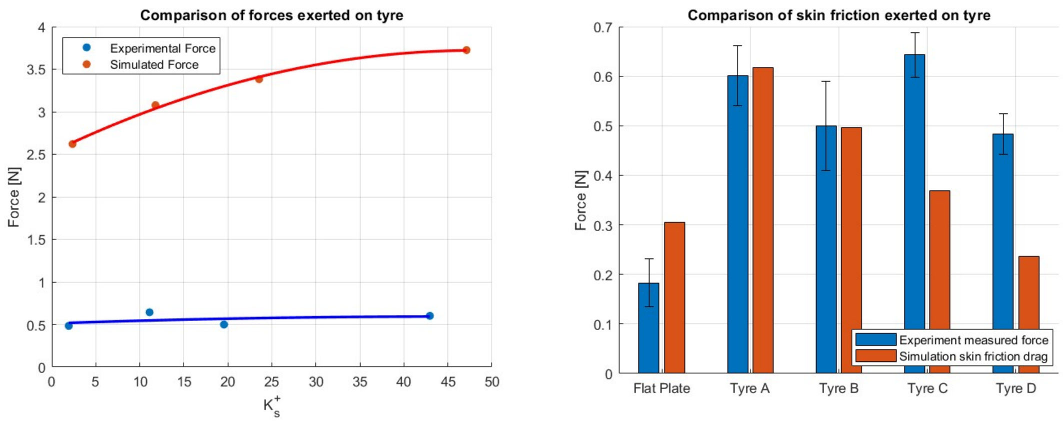


FIGURE 21: Results from Figure 19 at 64 [km/h]

Wall shear stress is needed to obtain roughness Reynolds number k_s^+ and is calculated using the measured force by $\tau_w = F_w/S$. The simulation and experimental results are then compared by plotting the force against k_s^+ (Figure 22a). The observation is made that the CFD simulation estimates a total force in the range of 2.6 - 3.7 [N], whereas the experiment ranges from only 0.483 - 0.643 [N]. This difference is so significant that the experiment results are closer to the simulated skin friction drag than the total. To investigate this difference, the measured force is compared to the simulated skin friction drag (Figure 22b). A better agreement between these results is seen, specifically for tyres A and B, where the simulated force falls within the error bars from the experiment. The flat plat error bars do not encompass the simulated value by approximately 74 [mN], making it remarkably difficult to identify its cause. On the other hand, tyres C and D show no correlation with the CFD simulation, which could be for several reasons. Firstly, an inaccurate measurement of the tyre's roughness was made by a vernier calliper, which is not an ideal tool for measuring roughness height. Secondly, the sand grain approach used in the CFD assumes that the topology of the tyre resembles sphere-like sand grains, which is not the case (Figure 8). These factors would affect the k_s^+ values specific to the tyre's roughness, potentially causing this result.



(A) (B)

FIGURE 22: Comparison of force from the two methods (a) Simulated drag and measured force (b) Simulated skin friction drag and measured force

5.2 Limitations of study

The major drawbacks of this study are uncertainties and errors, which prevent a definitive conclusion from being drawn.

The effectiveness of the CFD simulation for a comparative purpose is limited by the method with which the roughness parameter k_s was chosen. As stated in section 2, the topology of the tyre was simplified with the use of the sand-grain equivalent, which assumes spherical surface protrusions. The actual tyre roughness (Figure 8) consists of ridges arranged in straight lines, circular patterns and other layouts. [2, p. 14] lists correlations proposed for defining different surface roughnesses empirically based on studies conducted in this area. In addition, measuring the roughness height k_s and the tyre's side profile was not conducted adequately. Regardless of how the tyre was replicated in the simulation, the choice of RANS and turbulence model also impacted the result. Turbulence models use different assumptions and estimations to predict how the flow detaches and re-attaches to the surface, leading to different pressure drag estimations.

On the other hand, the wind tunnel analysis is limited by uncertainties in the measurement method, which can lead to potential systematic errors. This issue largely causes the large error bars to overlap and prevent an adequate conclusion from being drawn. Furthermore, the experiment method chosen for this study cannot separate the measured force into pressure and skin friction components; instead, it relies on the assumption that pressure drag is insignificant. This tremendous limitation prevents the measurement setup from quantifying the actual skin friction drag, which is the main objective of this study.

5.3 Recommendations for future improvements

To improve this study, the CFD simulation should be extended to a more extensive range of velocities, allowing for a more detailed comparison and understanding of the flow behaviour. In addition, a rendered or computer-aided model of the tyre would benefit the issues with modelling the tyre's side profile. A roughness tester should also be used instead of a vernier calliper when measuring the roughness height of the tyre's surface. Furthermore, the CFD model could be improved by trying different turbulence models on a flat plate geometry and seeing how they can be tuned to resemble the flow over the tyre more closely.

The wind tunnel analysis would also benefit from insight into the flow behaviour. One option would be to implement HWA in tandem with the FE. Although HWA is an indirect measurement, its implementation would provide insight into the velocity profile, which could be directly compared with the CFD simulation for a more rewarding analysis like determining the flow regimes established in section 2. Further research into the uncertainty caused by misalignment and the effects of the gap on force measurements would be beneficial for the reproducibility of this experiment.

6 Conclusion

In this paper, a FE design constructed to measure the skin friction drag of a tyre is proposed along with a CFD simulation used for comparative analysis. Unlike traditional arrangements, four leaf springs were used to suspend the FE at a height that ensures the tyre surface is flush with the wind tunnel while also constraining the FE's movement to a single DOF in the streamwise direction. A typical uncertainty error of the load cell, along with a high SNR caused by the vibrations, leads to a standard deviation error upwards of 41 [mN] for a windspeed of 64 [km/h]. Such deviation results in overlapping error bars that prevent drawing a definitive conclusion from the data. Additionally, a CFD simulation implementing RANS was tasked with replicating the scenario. Modelling the surface roughness using a sand-grain equivalent height of the tyre's topology revealed significantly larger drag forces across the board than measured with the setup.

Measurements of skin friction drag for a flat plate showed a 74 [mN] difference with simulated values, indicating poor agreement in the data. A detailed uncertainty analysis revealed that the most influential error contributions are systematic and cause issues with the experiment's reproducibility. Calibration of the FE's sensor requires known forces, which are necessary to improve the result's accuracy. This experiment further considered the flow over tyre surfaces varying in surface roughness. The measured skin friction drag for two of the four tyres showed favourable agreement with the friction component of the CFD simulation. Along with a smoke test conducted during the experiment, the results indicate issues with the simulation's method of quantifying surface roughness and inaccurate modelling of the tyre's side profile.

The findings of this study suggest that a FE set-up cannot adequately quantify the difference in skin friction drag caused by varying surface roughnesses. However, the analysis has been significantly constrained by substantial uncertainties and modelling errors inherent in the approach. The standard deviation and uncertainty errors of the experiment, coupled with the inaccurate modelling of the tyre's surface and side profile prompt further research in the accuracy and reliability of the FE design.

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